H9T Developer Edition: Engineering Evaluation

Reproducibility, interfaces, implementation boundaries, benchmark design, and validation roadmap

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Implementation summary

Reproduction: unzip this release, install numpy, scipy, scikit-learn, scikit-image, and run `OPENBLAS_NUM_THREADS=1 python followup_v04.py`. The CSV `followup_results.csv` is regenerated from deterministic seeds. The code contains its own nine-point matrix definition and assertions for inverse reconstruction.

Why software engineers should care

H9T is a compact, deterministic, separable, invertible nine-point operator. It can be useful as an interchangeable feature-engineering or hierarchical indexing primitive, provided a target application justifies it. Current image compression and anomaly benchmarks do not establish an advantage over DCT-II.

API contract

Inputs: finite floating-point arrays or 9-point vectors. Outputs: transform coefficients. Inverse must restore inputs to floating-point precision; lossy reconstruction is a separate quantizer/codec layer. Coefficient-selection masks must be stored, not silently discarded. Do not present higher precision as a speed improvement.

Reproducible commands

Run `python followup_v04.py` in the same folder as the script to write followup_results.csv. The environment must supply supported NumPy/SciPy/scikit-learn/scikit-image versions. Full version locking and CI are still required before a formal software release.

Scope and research questions

We investigate the normalized nine-point Helmert contrast transform (H9T) as a reversible numerical operator and explore whether its coefficient representation benefits image coding and classification. The tests address reconstruction, practical compressed-byte accounting, and held-out anomaly detection. This is a preliminary method study, not proof of novelty, production-grade compression, or general superiority.

Mathematical definition

Let H be the 9 by 9 matrix whose first row is 1/3 in all entries. For k = 1,...,8, row k has its first k entries equal to 1/sqrt(k(k+1)), entry k+1 equal to -k/sqrt(k(k+1)), and remaining entries zero (one-based row description). Then H H^T = I analytically and inverse reconstruction is x = H^T(Hx). The name H9T denotes the nine-point packaging and testing of this established Helmert-type basis. We do not claim that the basis is a new orthogonal transform.

Data and codec protocol

For imaging, three example images from scikit-image (camera, coins, astronaut converted to grayscale) were cropped to 252 by 252 pixels and normalized to [0,1]. H9T and DCT-II used independent 9 by 9 blocks. A two-level orthonormal Haar wavelet transform operated on the whole 252 by 252 field. For each transform, 5%, 10%, or 20% of global coefficients with greatest absolute value were retained. Retained coefficients were uniformly quantized in increments of 0.002, represented as signed 16-bit integers, accompanied by a packed binary coefficient mask, and compressed with zlib level 9. Reported bpp means compressed payload bytes times eight divided by pixel count. Headers, container

metadata, and application overhead were not included; this is a prototype coefficient-stream measurement, not a standardized image codec.

Image rate-distortion result

The table reports PSNR in dB followed by compressed coefficient-stream bpp at the same 10% retained-coefficient target. The actual bpp differs across methods. Thus the table does not establish which method wins at fixed bitrate: bitrate-matched interpolation or codec tuning is still required.

Image	H9T PSNR / bpp	DCT-II PSNR / bpp	Haar-L2 PSNR / bpp
camera	37.70 / 1.439	41.67 / 1.472	37.65 / 1.353
coins	29.57 / 1.555	31.34 / 1.581	29.71 / 1.491
astronaut	30.64 / 1.545	36.47 / 1.595	31.70 / 1.527

Anomaly detection protocol

Synthetic nine-sample time windows combine randomized sinusoids and Gaussian noise. Half are labelled anomalous, containing a single spike, a three-sample burst or a linear drift. For each of three independent seeds (11, 23, 37), 2,400 training windows and 1,600 held-out windows were generated separately. A histogram gradient boosting classifier with identical hyperparameters was trained on raw samples, H9T coefficients, DCT-II coefficients, FFT magnitude, or a reversible Haar transform of the first eight samples plus the ninth sample. Reported figures are the mean of three held-out evaluations. No real sensor datasets or confidence intervals from independent physical sites are included.

Anomaly classification result

Across three synthetic seeds, H9T performed better than raw nine-point data with the chosen classifier, but DCT-II performed better than H9T; there is no basis for an H9T-superiority claim on this experiment. Because all invertible transforms retain information, apparent predictive gains are model/regularization/representation dependent; identical well-tuned models and test sets are essential.

Representation	Mean ROC-AUC	Mean average precision
Raw	0.8351	0.8576
H9T	0.8594	0.8806
DCT-II	0.9187	0.9255
FFT-magnitude	0.8788	0.8905
Haar	0.8907	0.9071

Internal audit and corrected benchmark

During this follow-up a matrix-index ordering error was found in the earlier v0.3 two-dimensional inverse routine. It could invalidate its prior image quality figures. The v0.4 implementation uses the correct inverse indexing and enforces full-coefficient round-trip tolerances below 2e-13 before lossy testing. The corrected v0.4 values supersede v0.3 image results. Historic seven-level float64 verification (approximately 5.6e-16 relative residual) is a different test and was not rerun here.

Limitations and falsification conditions

We have not demonstrated superior bitrate-matched image coding, superior industrial anomaly detection, real-world robustness, trained deep-learning enhancement, statistically significant wins, or better CPU/GPU efficiency. The Haar comparator differs in block structure from the 9x9 transforms. There are only three sample images, three synthetic seeds, and one model family. Coefficient-stream bpp is not an independently decodable media file including full metadata. An independent reviewer should require fixed-bpp sweeps, real industrial datasets, uncertainty intervals, transform-time/peak-RAM accounting, side-information costs, cross-site splits, and ablation tests.

Next validation gates

Gate A: equal-bit-rate PSNR/SSIM curves over at least 50 images and multiple codecs, including JPEG and wavelet baselines. Gate B: quantized entropy-coded decodable bitstreams with all metadata included. Gate C: public real sensor time-series with site-held-out or time-held-out labels, multiple classifiers, bootstrap confidence intervals, PR-AUC, false alarms per hour and latency. Gate D: practical advantage under matched conditions or explicitly document a negative result.

Source and credit

Supporting source: H9T Developer Edition verification and initial application research packages supplied within this project. External algorithmic background: classical Helmert orthogonal contrasts, discrete cosine transforms, Haar wavelets, PSNR, ROC-AUC and zlib. Any scholarly release should add complete bibliographic citations and an explicit versioned software archive DOI.

Conclusions

Developer conclusion: the prototype is reproducible and technically usable for exploratory work; the matched studies favor DCT-II on these datasets. A release should describe limitations as prominently as capabilities.